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## H2 Biology

9744/02

Paper 2 Structured Questions

15 September 2021

2 hours

Candidates answer on the Question Paper.  
No Additional Materials are required.

### READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |     |
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| 1                  |     |
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| Total              | 100 |

This document consists of **30** printed pages and **2** blank pages.

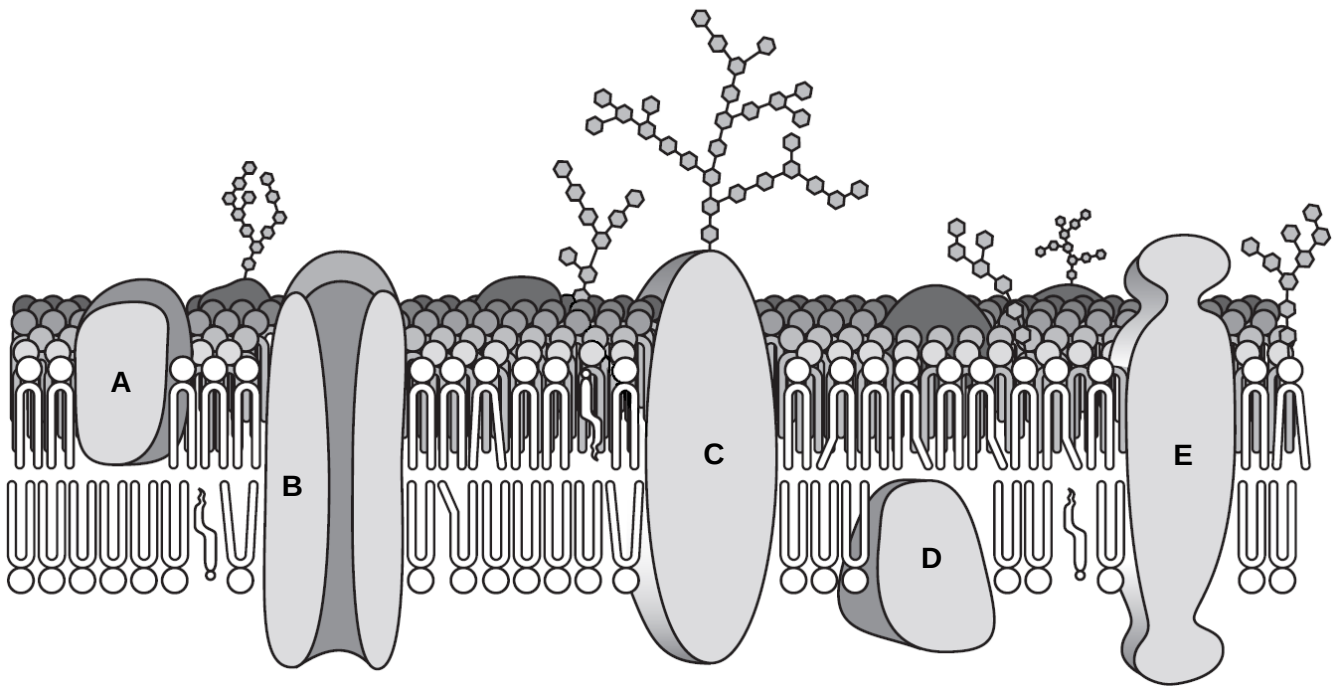
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Answer **all** questions.

- 1** Hormones are responsible for regulating the permeability of kidney cells to water and dissolved ions.

Kidney cells in the collecting duct are impermeable to water. To allow water molecules to move into cells, the hormone vasopressin upregulates the expression of aquaporin channel proteins.

Fig. 1.1 shows the cell surface membrane of a kidney cell, with five proteins labelled **A** to **E**.



**Fig. 1.1**

**(a)** From Fig. 1.1,

- (i)** identify which protein represents the structure of an aquaporin.

.....[1]

- (ii)** draw an arrow on Fig. 1.1 to show the direction of movement of water molecules. [1]

- (iii)** explain how the arrangement of molecules supports the fluid mosaic model.

.....  
 .....  
 .....  
 .....[2]

Vasopressin receptors are G-protein coupled receptors with a transmembrane domain.

**(b)** Explain how vasopressin receptors are held in the membrane.

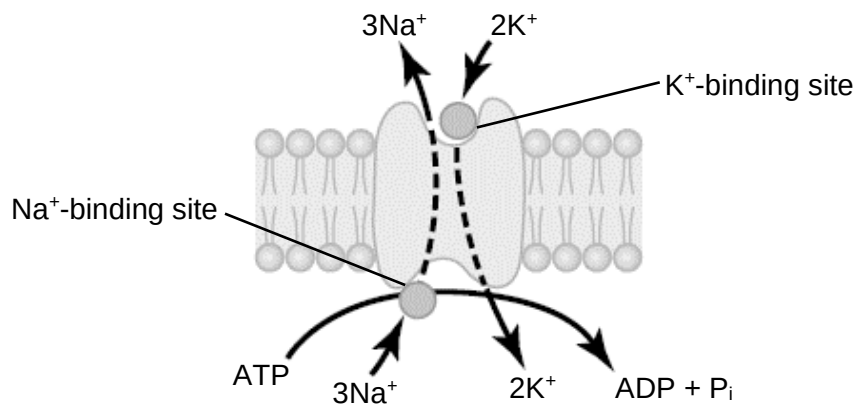
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..... [2]

The lipid-soluble hormone aldosterone regulates the level of sodium and potassium ions. It achieves this by upregulating the expression of  $\text{Na}^+\text{-K}^+$  pumps. Fig. 1.2 shows how these protein pumps facilitate the transport of sodium ions into the cell and potassium ions out of the cell.



**Fig. 1.2**

**(c)** Explain why  $\text{Na}^+\text{-K}^+$  pumps allow  $\text{Na}^+$  and  $\text{K}^+$  to move across the cell surface membrane.

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..... [2]

**(d)** Describe **one** difference in the transport of sodium ions and water molecules into kidney cells in response to aldosterone and vasopressin respectively.

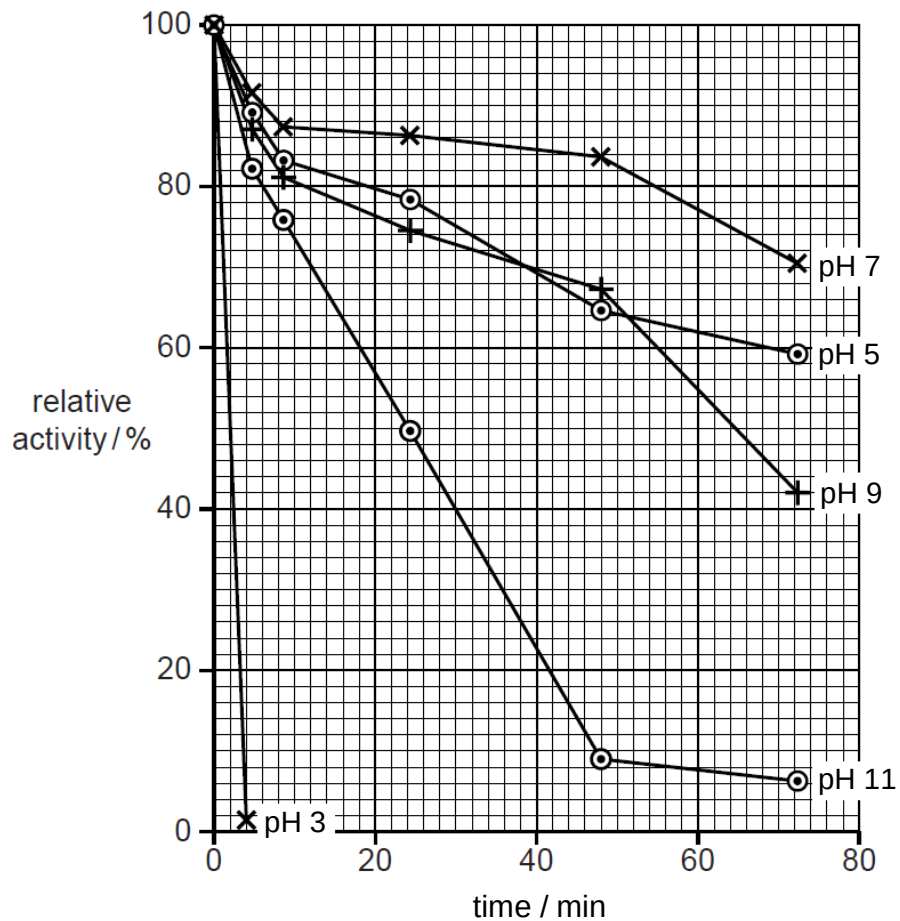
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..... [1]

[Total: 9]

- 2 Tuberculosis (TB) is an infectious disease caused by the pathogenic bacterium *Mycobacterium tuberculosis*. In this bacterial species, an enzyme known as fructose 1,6-bisphosphate aldolase (FBA) is critical for glucose metabolism.

Fig. 2.1 shows the relative activity of *M. tuberculosis* FBA measured at different pH levels.



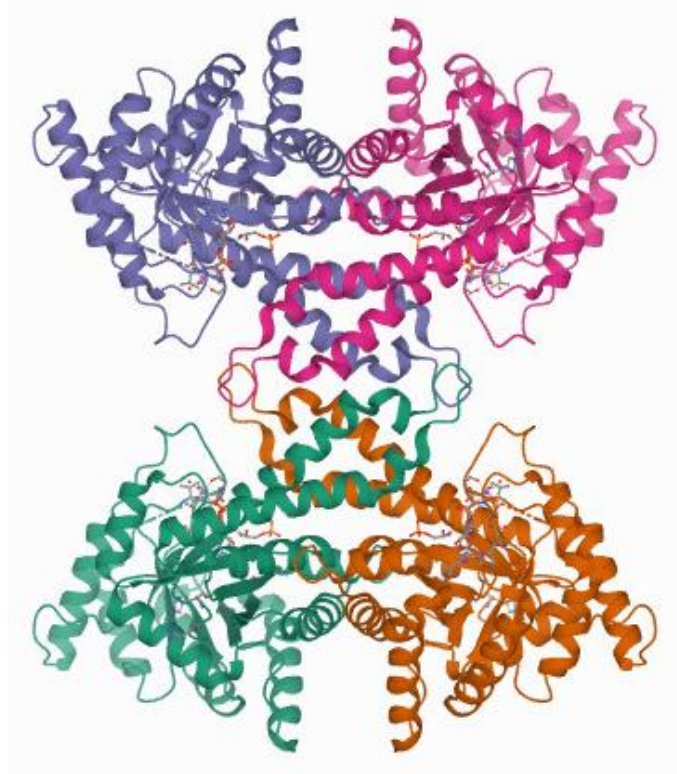
**Fig. 2.1**

- (a) With reference to Fig. 2.1, distinguish between the effect of neutral and alkaline pH on the activity of FBA.

.....

..... [1]

Fig. 2.2 shows the structure of FBA, which is made of four identical polypeptide subunits.



**Fig. 2.2**

**(b) (i)** On Fig. 2.2, label and name a secondary structure.

[1]

**(ii)** Explain how changing the pH level affects the structure and activity of FBA.

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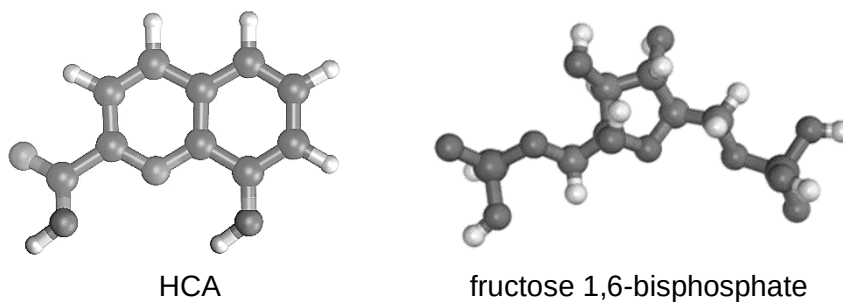
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.....[4]

Recent research has identified a potential therapeutic drug that targets the FBA enzyme known as 8-hydroxyquinoline carboxylic acid (HCA). Fig. 2.3 shows the molecular shapes of HCA and fructose 1,6-bisphosphate, which is the substrate for FBA.



**Fig. 2.3**

**(c)** Suggest how HCA may act as a therapeutic drug to treat tuberculosis.

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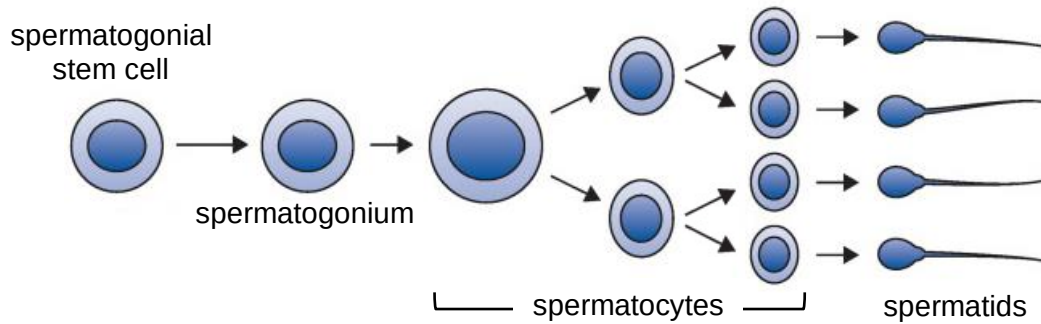
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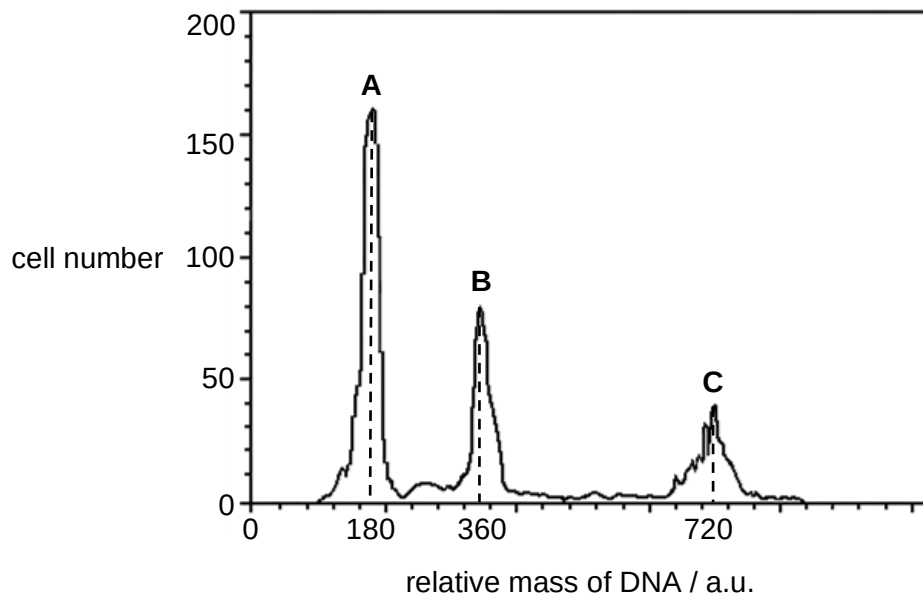
[Total: 10]

3 Fig. 3.1 shows the normal differentiation pathway of a spermatogonial stem cell into spermatids.



**Fig. 3.1**

Fig. 3.2 shows three groups of spermatocytes, **A**, **B** and **C**, in different stages of cell division. The peak relative mass of DNA is indicated for each group of cells.



**Fig. 3.2**

(a) With reference to Fig. 3.2,

(i) identify the group of spermatocytes that are formed at the end of

[1]

| meiosis I | meiosis II |
|-----------|------------|
|           |            |

(ii) explain your answer in (a)(i).

.....

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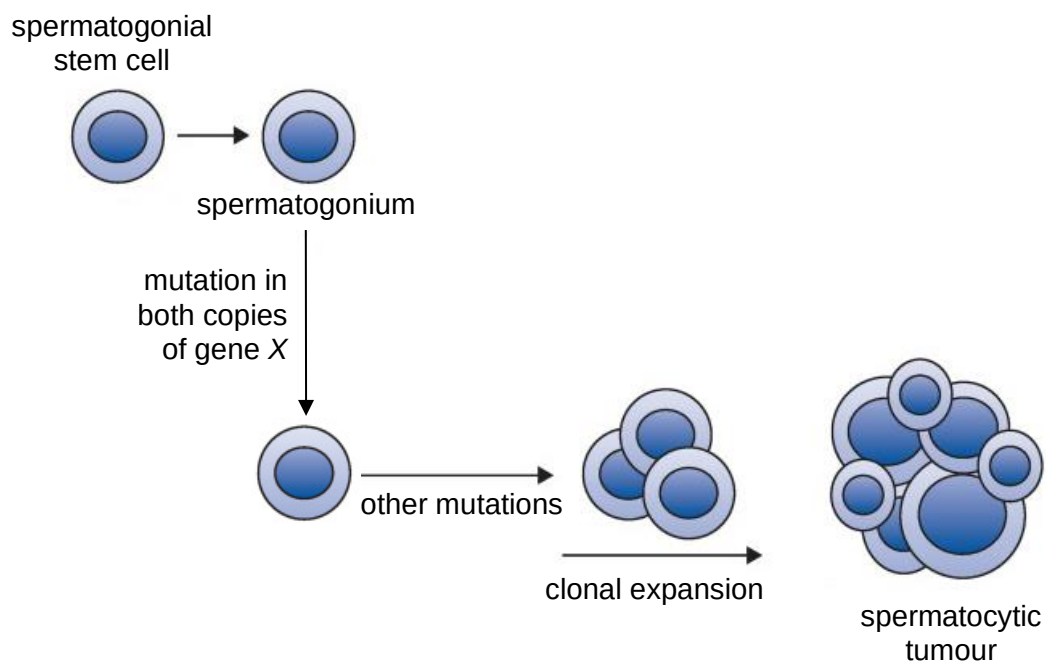
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.....[3]

Germ cell tumours result from uncontrolled division of germ cells, which give rise to gametes. In males between the ages of 15 to 35 years, the risk of testicular cancer increases with age.

Fig. 3.3 shows how a tumour may result from uncontrolled cell division of a spermatogonium (male germ cell), instead of normal differentiation into spermatids.



**Fig. 3.3**

(b) (i) Suggest the name of gene X.

.....[1]

(ii) Explain how gene *X* may result in uncontrolled cell division of a spermatogonium.

.....

.....

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.....

.....

.....[3]

(c) Cancer treatments, including chemotherapy and radiotherapy, can often result in loss of fertility. In male cancer patients, sperm cryopreservation (freezing of sperm cells) is one method of preserving fertility.

Using your knowledge of stem cells, suggest why fully differentiated sperm cells are preserved instead of spermatogonial stem cells.

.....

..... [1]

[Total: 9]

- 4 In the bacterium *Thermus thermophilus*, a sequence of DNA contains two overlapping genes, *rnpA* and *rpmH*, which code for RNase P and a ribosomal protein respectively.

Fig. 4.1 shows the first 30 bases in the DNA sequence. Following process X, a different strand of nucleic acid is formed, and it contains the translation start sites for RNase P and the ribosomal protein.

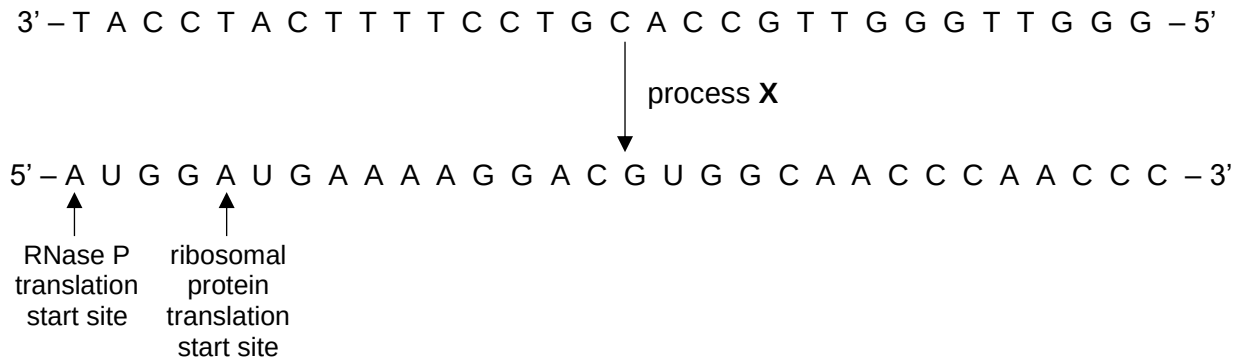


Fig. 4.1

- (a) (i) Briefly describe process X.

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.....[3]

- (ii) Suggest why the product of process X can code for two different polypeptide products.

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.....[2]

In eukaryotic cells, the *FECH* gene codes for ferrochetalase, which is the final enzyme in the metabolic pathway of haem synthesis.

A study done in zebrafish found that a point mutation in the *FECH* gene resulted in a disease phenotype known as dracula. The disease is characterised by lysis of red blood cells upon exposure to light.

Fig. 4.2 shows a segment of the *FECH* template DNA sequence in normal and dracula zebrafish, and Table 4.1 shows the genetic code.

normal

|     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|
| 27  | 28  | 29  | 30  | 31  | 32  | 33  |
| CAC | GTG | GAC | AGT | GGA | CTC | ATC |

dracula

|     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|
| 27  | 28  | 29  | 30  | 31  | 32  | 33  |
| CAC | GTG | GAC | ATT | GGA | CTC | ATC |

**Fig. 4.2**

**Table 4.1**

|   | U   |     | C   |     | A   |      | G   |      |                  |
|---|-----|-----|-----|-----|-----|------|-----|------|------------------|
| U | UUU | Phe | UCU | Ser | UAU | Tyr  | UGU | Cys  | U<br>C<br>A<br>G |
|   | UUC |     | UCC |     | UAC |      | UGC |      |                  |
|   | UUA | Leu | UCA |     | UAA | STOP | UGA | STOP |                  |
|   | UUG |     | UCG |     | UAG | STOP | UGG | Trp  |                  |
| C | CUU | Leu | CCU | Pro | CAU | His  | CGU | Arg  | U<br>C<br>A<br>G |
|   | CUC |     | CCC |     | CAC |      | CGC |      |                  |
|   | CUA |     | CCA |     | CAA | Gln  | CGA |      |                  |
|   | CUG |     | CCG |     | CAG |      | CGG |      |                  |
| A | AUU | Ile | ACU | Thr | AAU | Asn  | AGU | Ser  | U<br>C<br>A<br>G |
|   | AUC |     | ACC |     | AAC |      | AGC |      |                  |
|   | AUA | Met | ACA |     | AAA | Lys  | AGA | Arg  |                  |
|   | AUG |     | ACG |     | AAG |      | AGG |      |                  |
| G | GUU | Val | GCU | Ala | GAU | Asp  | GGU | Gly  | U<br>C<br>A<br>G |
|   | GUC |     | GCC |     | GAC |      | GGC |      |                  |
|   | GUA |     | GCA |     | GAA | Glu  | GGA |      |                  |
|   | GUG |     | GCG |     | GAG |      | GGG |      |                  |

**(b)** Using Fig. 4.2 and Table 4.1,

**(i)** State the amino acid sequence for this segment of the normal polypeptide.

.....[1]

**(ii)** Explain how the point mutation results in non-functional ferrochetalase.

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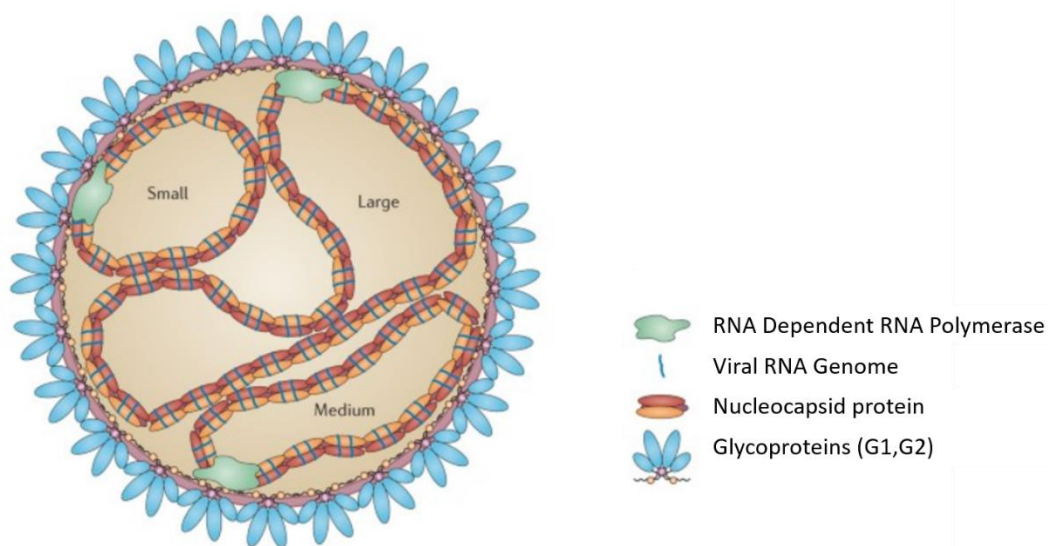
.....[2]

[Total: 8]

- 5 Hantaviruses are a group of viruses that normally infect rodents, but do not cause disease in them.

Humans may become infected with Hantaviruses through contact with rodent urine, saliva, or faeces. Some strains cause potentially fatal diseases, such as Hantavirus Haemorrhagic Fever with Renal Syndrome (HFRS), and Hantavirus Cardiopulmonary Syndrome (HCPS), in humans.

Fig. 5.1 shows the structure of a strain of Hantavirus. The genome of the virus consists of three segments – small, medium and large.



**Fig. 5.1**

- (a) (i) With reference to Fig. 5.1, describe **three** structural features which distinguish the Hantavirus from its human host cell.

.....

.....

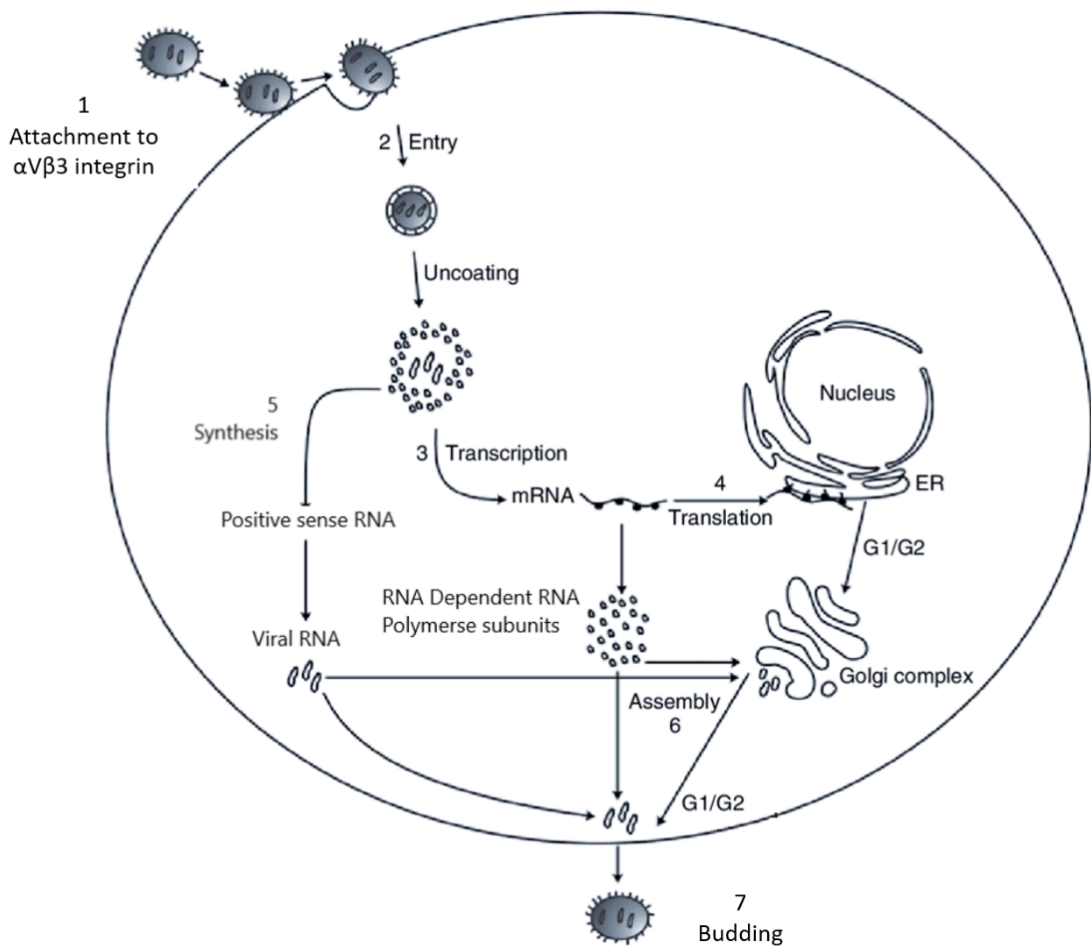
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.....[3]

Hantavirus replication takes place in macrophages and vascular endothelial cells, especially in the lung and the kidney. Fig. 5.2 shows the reproductive cycle of the Hantavirus.



**Fig. 5.2**

- (ii) With reference to Fig. 5.2, name another virus which has a similar reproductive cycle to the Hantavirus.

.....[1]

- (iii) Describe how the reproduction of the Hantavirus is similar to the virus named in (a)(ii).

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.....[3]

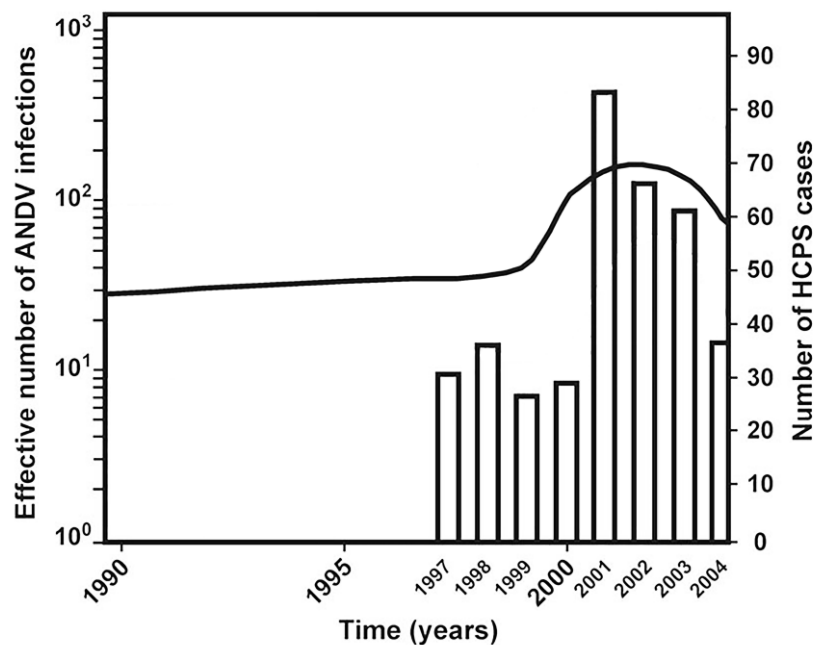
- (b) Human Hantavirus infections are assumed to occur accidentally, and humans represent a dead end host for the Hantavirus. Transfer of virus particles from infected to uninfected people normally does not occur.

One exception is the Andes Hantavirus (ANDV) strain found in South America, for which sporadic person-to-person transmissions have been reported. This finding reveals a worrying risk potential of Hantaviruses for human health.

- (i) Describe how the ANDV strain becomes capable of human-to-human transmission.

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 .....  
 .....[2]

Fig. 5.3 shows the data for ANDV infections, as reported by the Chilean Ministry of Health. The solid line represents the effective number of ANDV infections in the population. The bar graph represents the total number of human Hantavirus Cardiopulmonary Syndrome (HCPS) cases per year.



**Fig. 5.3**

- (ii) Suggest why there are no human HCPS cases reported before the year 1997.

.....  
 .....[1]

[Total: 10]

6 Fig. 6.1 illustrates the *lac* operon and its regulatory gene in *E. coli*.

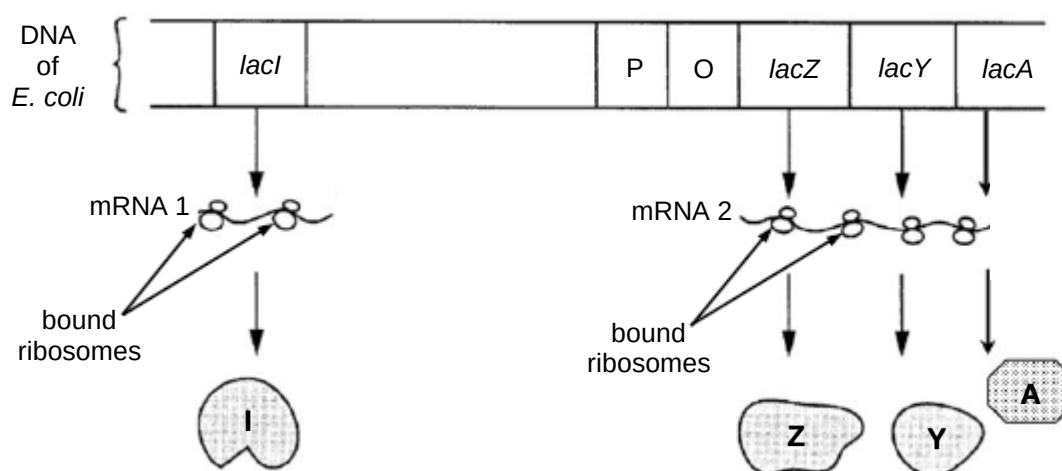


Fig. 6.1

- (a) With reference to Fig. 6.1, explain why ribosomes bound to mRNA 1 produce the same polypeptide but those bound to mRNA 2 produce three different polypeptides.

.....

.....

.....

..... [2]

- (b) A certain bacteriophage is known to insert its own genome within the *lacZ* gene of *E. coli*. In an experiment, *E. coli* cells were infected with the bacteriophages and the infected cells grown in a medium containing equal concentrations of glucose and lactose.

Fig. 6.2 shows how the number of infected *E. coli* cells varies with concentration of glucose and lactose.

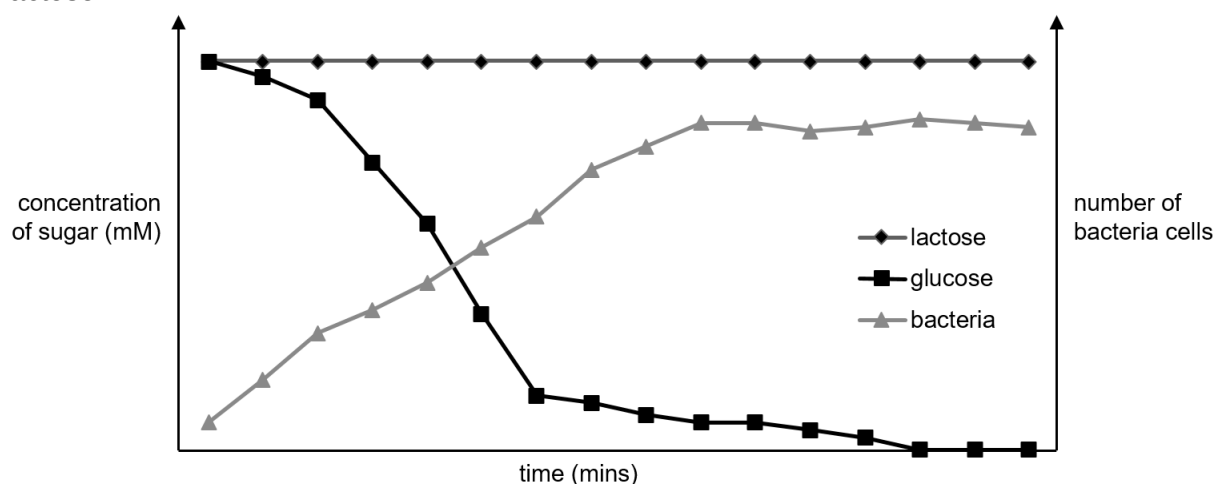


Fig. 6.2

With reference to Fig. 6.2, explain the trend observed in each of the three graphs.

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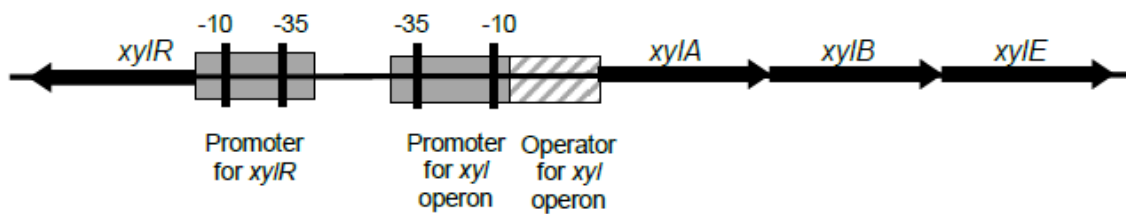
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- (c) The *xyl* operon contains genes involved in the breakdown of the sugar xylose. Fig. 6.3 shows the organisation of the *xyl* operon in another species of bacterium, *T. halophila*.

The *xyl* operon contains three structural genes (*xylA*, *xylB* and *xylE*) and is regulated by the protein product of the regulatory gene, *xylR*. The arrows represent the direction of transcription of the named genes.



**Fig. 6.3**

- (i) State whether the *xyl* operon is an inducible or repressible operon.

.....[1]

- (ii) Suggest how the *xylR* gene regulates the expression of the *xyl* operon when xylose is present.

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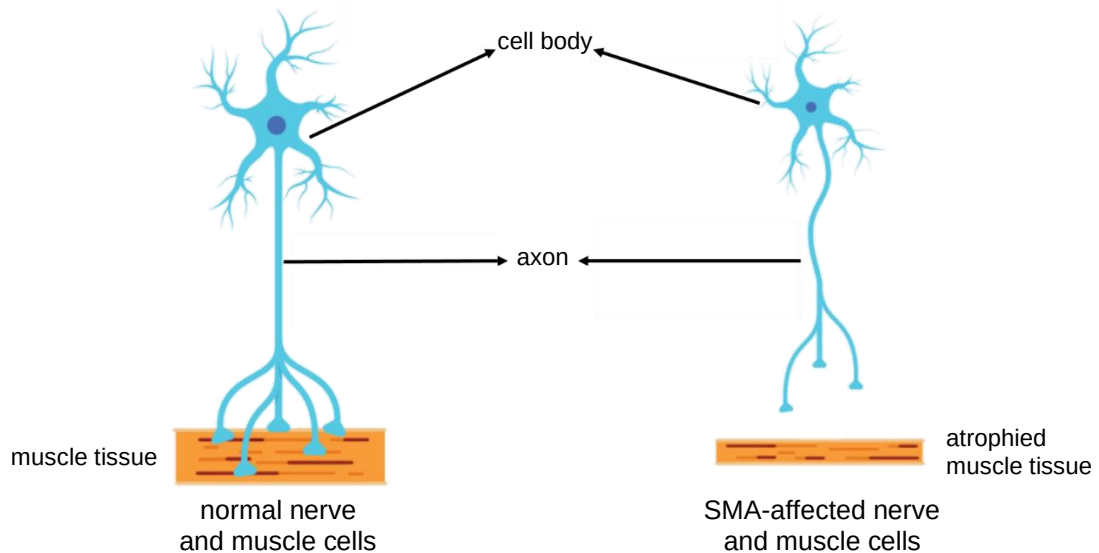
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.....[4]

[Total: 10]

- 7 Chromosome 5-related Spinal Muscular Atrophy (SMA) is caused by a mutation in the Survival Motor Neuron 1 (SMN1) gene. This gene is responsible for producing the Survival Motor Neuron (SMN) protein, which maintains the health and normal function of motor neurons.

In individuals with SMA, both copies of the SMN1 gene are mutated, leading to decreased production of SMN protein. Without proper levels of SMN protein, motor neurons in the spinal cord will be lost, preventing the muscles from receiving proper signals from the brain as shown in Fig. 7.1.



**Fig. 7.1**

- (a) Explain if the SMN1 mutation resulting in SMA is dominant or recessive in nature.

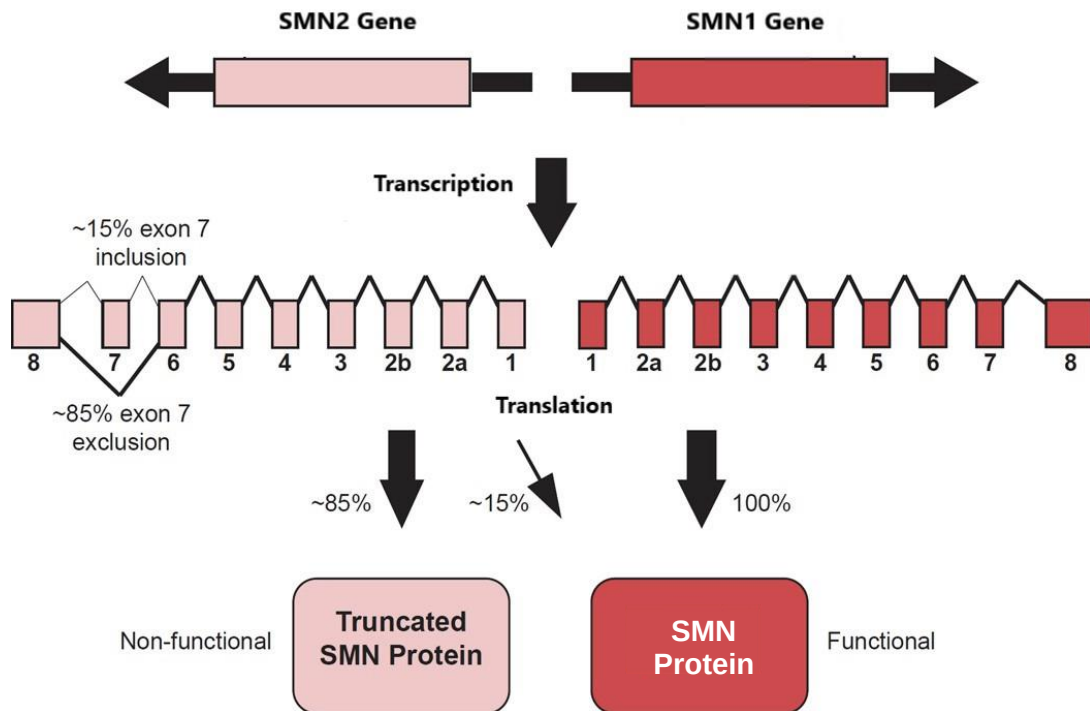
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..... [2]

- (b) A neighbouring gene on chromosome 5, called SMN2, also produces SMN protein. Most of the protein expressed by SMN2 genes is not functional, but a small percentage, around 15% are functional as illustrated by Fig. 7.2.



**Fig. 7.2**

- (i) With reference to Fig. 7.2, explain why 85% of the SMN proteins expressed from the SMN2 gene are synthesised as non-functional in nature.

.....[3]

Individuals can have multiple copies of the SMN2 gene. Normally, the number varies between zero and eight copies.

In the chromosome 5-related form of SMA, having three or more copies of the SMN2 gene is associated with a less severe disease manifestation.

- (ii) State the type of genomic modification which results in multiple copies of SMN2 gene in an individual.

.....[1]

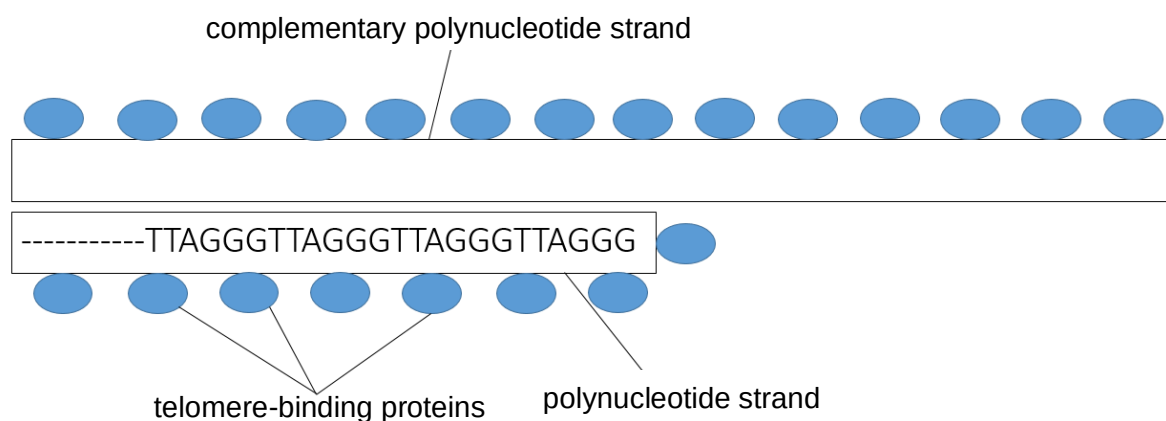
- (iii) Suggest how having higher copies of SMN2 gene results in less severe disease manifestation.

.....  
 .....[1]

- (c) Telomeres are found at the end of eukaryotic chromosomes.

Telomeres consist of a nucleotide sequence that is usually repeated about 3,000 times. Telomere-binding proteins are also found to be attached to these telomeric sequences.

Fig. 7.3 shows part of a telomere found in eukaryotic chromosomes.



**Fig. 7.3**

- (i) With reference to Fig. 7.3, state the nucleotide sequence which is repeated in the complementary polynucleotide strand.

.....[1]

- (ii) Suggest the role of telomere-binding proteins.

.....  
 .....[1]

- (iii) In the past, non-coding sequences were referred to as “junk DNA” which have no purpose in the genome.

Explain why the term “junk DNA” is misleading in the context of the telomere.

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 .....  
 .....[2]

[Total: 11]

**[Turn over]**

- 8 In the presence of light, potatoes produce a greening response. This results in the potato producing broad green leaves, short sturdy stems and long roots as shown in Fig. 8.1.



Fig. 8.1

Upon reception of light by the phytochrome receptor, two signal transduction pathways are elicited in a potato cell as shown in Fig. 8.2. Both pathways eventually result in the synthesis of greening response proteins such as photosynthetic enzymes as well as enzymes involved in chlorophyll production.

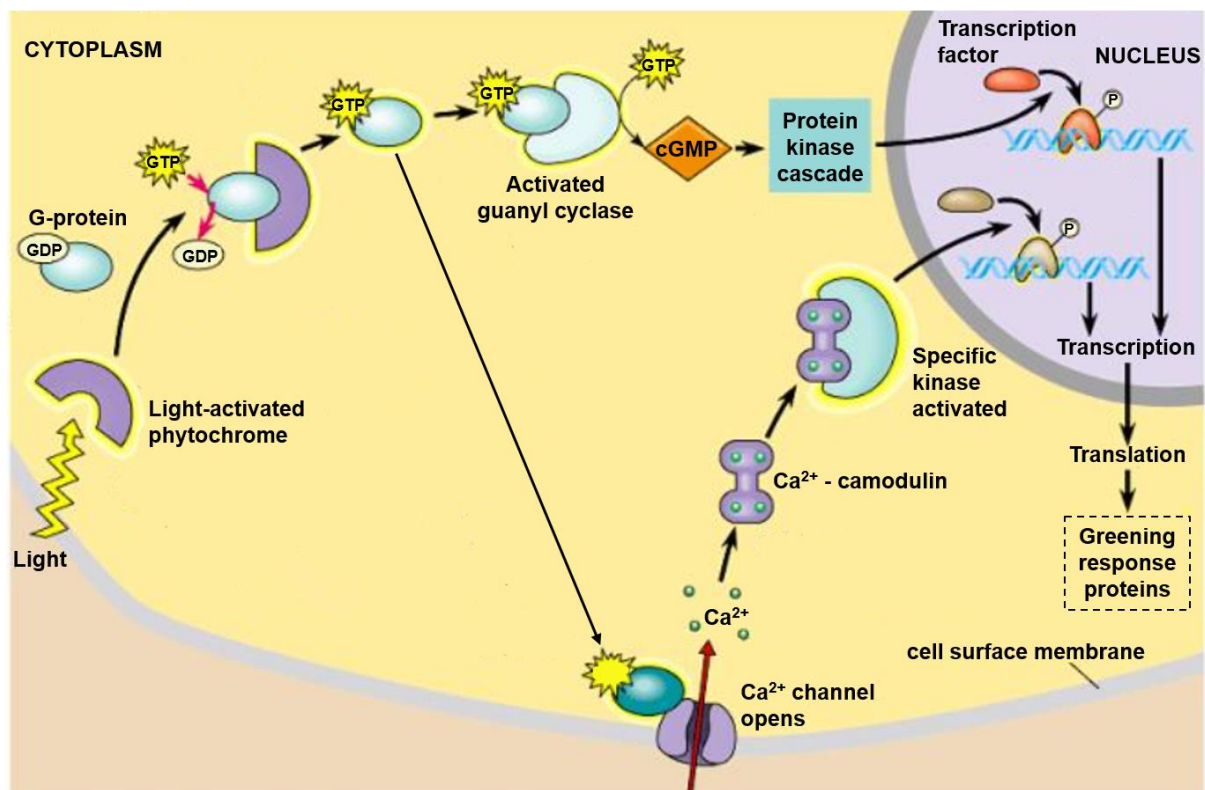


Fig. 8.2

- (a) Describe the role of second messengers.

.....

.....

.....

[2]

(b) With reference to Fig. 8.2,

(i) identify the second messengers involved in the signalling pathway.

.....[1]

(ii) explain **one** step of the signal transduction stage in the cytoplasm that illustrates signal amplification.

.....

.....

.....

.....[2]

(iii) compare the activation of calmodulin with that of the final protein kinase in a protein kinase cascade.

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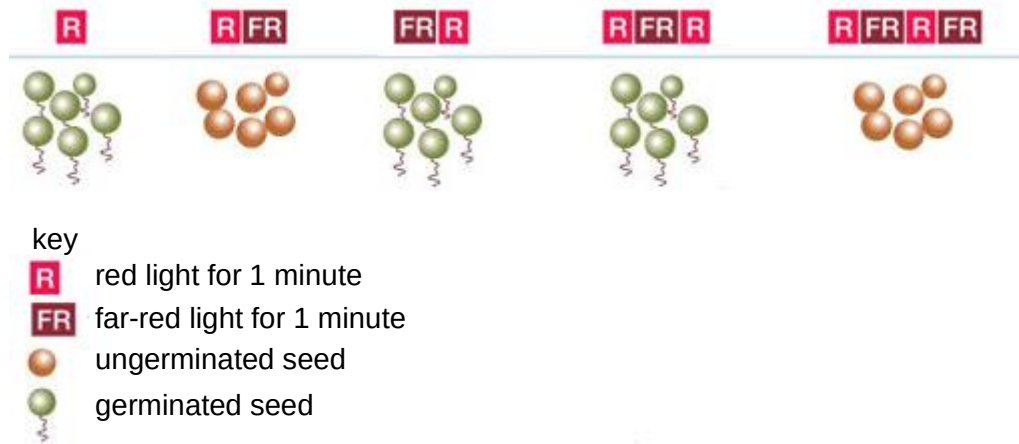
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.....[3]

- (c) Phytochromes are able to detect red light and far-red light.

An experiment was conducted to investigate the role of phytochromes and germination of lettuce seeds. The lettuce seeds were exposed to alternating periods of red light and far-red light for 1 minute each.

The results of the investigation are shown in Fig. 8.3.



**Fig. 8.3**

Explain which light, when detected by phytochromes, is more effective in inducing seed germination.

.....

.....

.....

..... [2]

[Total: 10]

- 9 Acute Immune Deficiency Syndrome can be caused by either of two retroviruses, known as Human Immunodeficiency Virus types 1 and 2 (HIV-1 and HIV-2).

Related viruses isolated from a number of other primate species are termed as Simian Immunodeficiency Virus (SIV).

Fig. 9.1 shows a phylogenetic tree illustrating the evolutionary relationships among the two major forms of HIV, HIV-1 and HIV-2, and SIV that afflicts non-human primates.

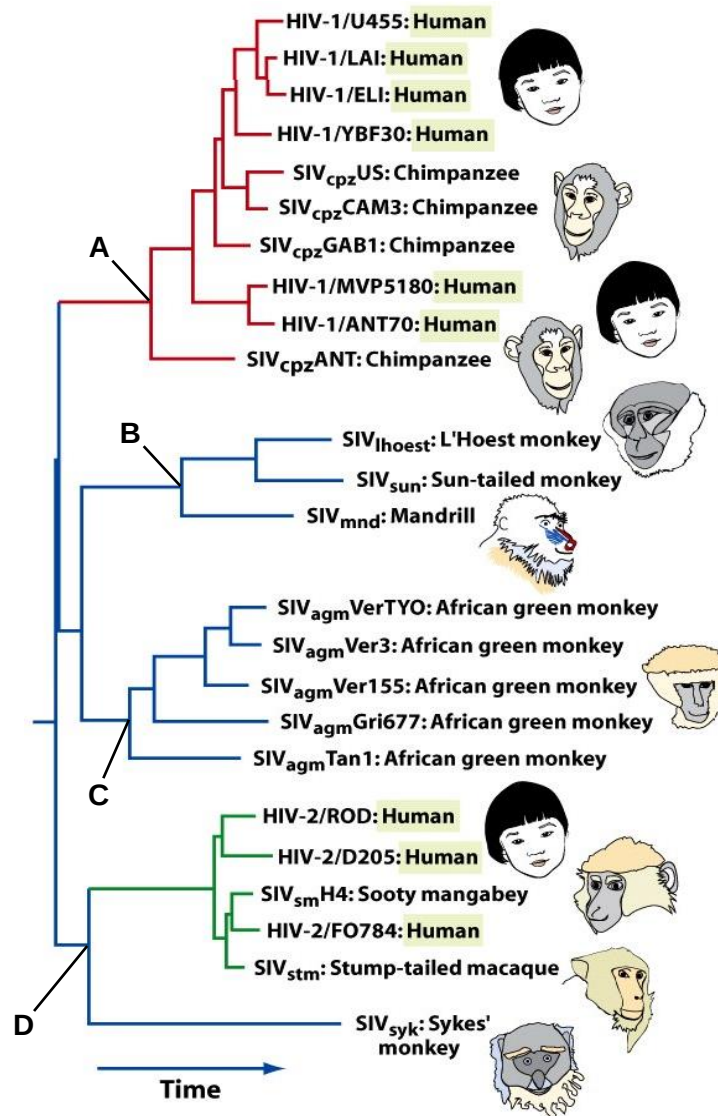


Fig. 9.1

- (a) Using the information given in Fig. 9.1,
- (i) briefly describe how the phylogenetic tree was constructed.

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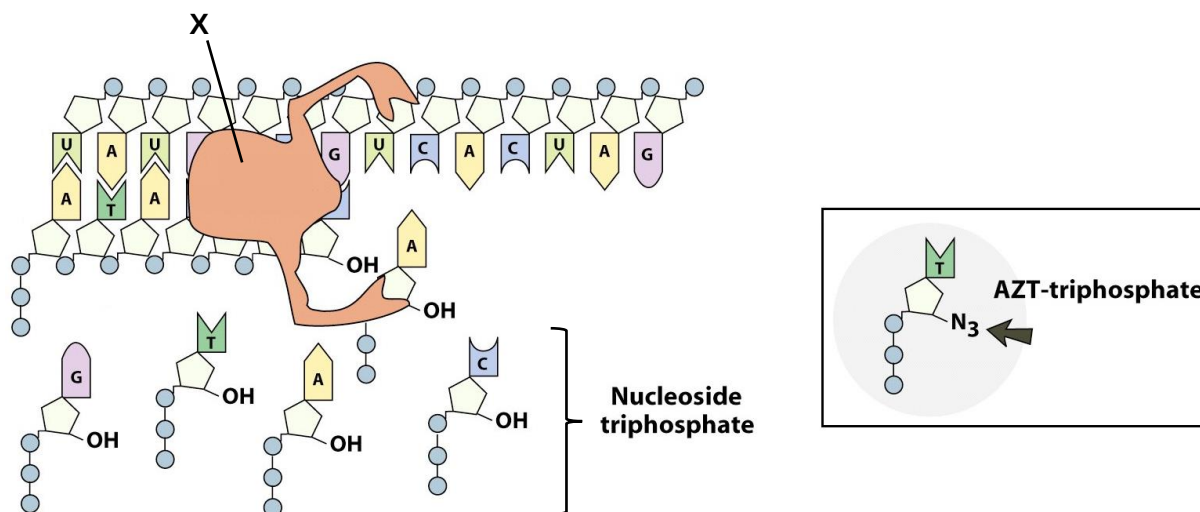
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.....[2]

- (ii) Nodes A, B, C and D have been labelled on Fig. 9.1 Using the information given, identify the **two** nodes which denote the common ancestors of HIV.

.....[1]

- (b)** Fig 9.2 shows a molecular process occurring within an HIV-infected host cell. An anti-viral drug AZT triphosphate was introduced into the cell and the process below was observed.



**Fig. 9.2**

- (i) Identify molecule **X** and briefly describe the reaction it is involved in as seen in Fig. 9.2.

.....[2]

- (ii)** Explain how AZT-triphosphate serves as an anti-viral drug.

.....[2]

- (c) Table 9.1 shows the percentage of AZT resistant viral strains in a group of patients who underwent AZT therapy. The percentage of AZT resistant viral strains was assessed at 1, 2, 12 and 18 months after the start of AZT therapy.

**Table 9.1**

| time after start of therapy / months | AZT resistant viral strains / % |
|--------------------------------------|---------------------------------|
| 1                                    | 2                               |
| 2                                    | 10                              |
| 12                                   | 20                              |
| 18                                   | 50                              |

With reference to Table 9.1, explain how the use of AZT therapy can lead to the evolution of AZT resistant viral strains.

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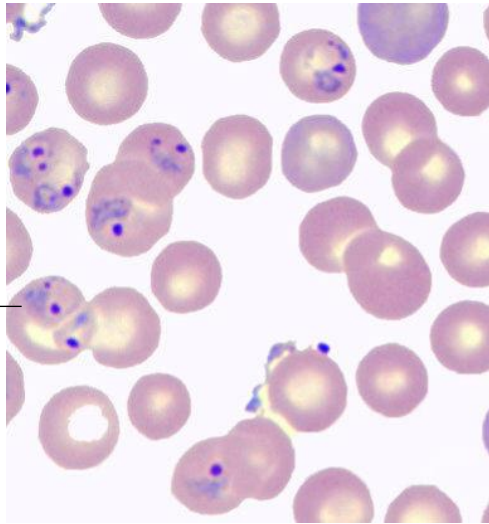
..... [4]

[Total: 11]

- 10** Malaria is a serious disease, thought to affect more than 220 million people around the world. Every year, over three quarters of a million people die from malaria, 90% of whom are in Africa.

Malaria is caused by the malarial parasite which infects red blood cells in the body, as shown in Fig. 10.1.

red blood cell  
infected with  
malarial parasite



**Fig. 10.1**

10 to 15 days upon initial infection with the malaria parasite, several symptoms including headache, fever and vomiting will manifest. The disease can rapidly affect many vital organs and become life threatening.

- (a)** After repeated exposure to a variety of malarial strains, some people develop a natural immunity to the disease.

Explain how a person could develop natural immunity to malaria.

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.....[4]

One way of producing immunity to malaria is through vaccination.

Randomised trials have taken place for a new vaccine called RTS,S. In 2011, 447 children aged 5 to 17 months from Kenya and Tanzania were vaccinated with RTS,S. The children were then monitored to see if they developed malaria. The results are shown in Table 10.1.

**Table 10.1**

| vaccination with RTS,S |                               |                                   |
|------------------------|-------------------------------|-----------------------------------|
| number of subjects     | number that developed malaria | percentage that developed malaria |
| 447                    | 38                            | 8.5                               |

The drug company that developed the RTS,S vaccine came to the conclusion that the vaccine had an effectiveness of 56% and suggested that it could be used as an effective vaccination against malaria in the future. Other researchers have suggested that alternative approaches to preventing malaria should be considered.

- (b)** Suggest a reason for and against the introduction of a large-scale vaccination programme using RTS,S.

*For*

.....  
 .....[1]

*Against*

.....  
 .....[1]

[Total: 6]

- 11** Tetanus is a very serious disease and can be fatal. Symptoms of the disease are caused by a toxin produced by *C. tetani*.

There are different ways to gain immunity to the disease, such as described in situations A and B shown in Table 11.1.

**Table 11.1**

| situation | description of immunity                                                                                                                                                       |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A         | A student has immunity after receiving three rounds of a vaccine, which consists of modified tetanus toxin.                                                                   |
| B         | An unvaccinated person, who has just been infected by the pathogen, gains immediate immunity by being given an injection containing antibodies specific to the tetanus toxin. |

- (a)** Compare the type of immunity described in situations A and B.

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 .....  
 .....  
 .....[2]

- (b)** *C. tetani* infection can be treated with antibiotics.

Explain if penicillin, which is an antibiotic, will be effective in the treatment of infections caused by other pathogens, such as viruses.

.....  
 .....  
 .....  
 .....[2]

Allergens are molecules that can cause allergic diseases. Allergens present on pollen produced by some species of *Bromus* can result in hayfever (allergic rhinitis) in some people.

Immunoglobulin E (IgE) is the class of antibody produced in response to allergens. IgE has a high affinity for Fc receptors on mast cells, which are a type of white blood cell.

Fig. 11.1 is an outline summary of how hayfever develops after an initial 'sensitisation' process.

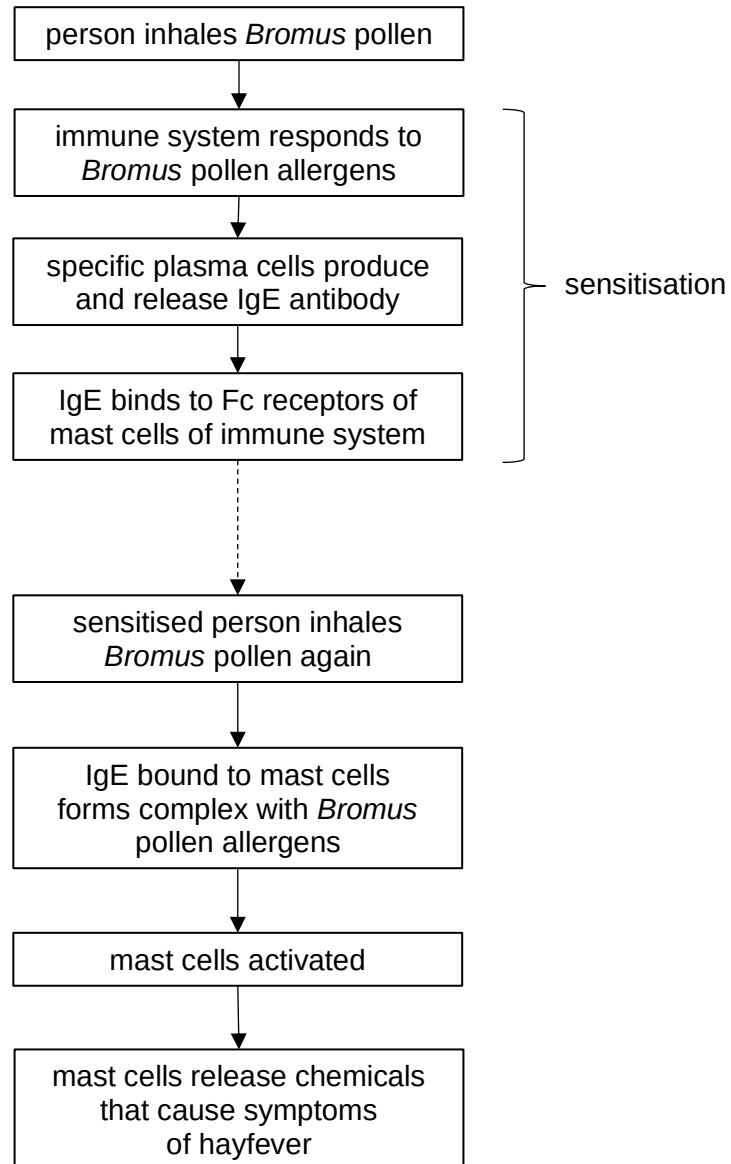


Fig. 11.1

- (c) Omalizumab is an antibody that binds to IgE molecules, and can be used in the treatment of hayfever.

Suggest how Omalizumab may act to reduce the symptoms of hayfever in a sensitised person.

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